

Отчёт о лабораторной работе

Лабораторная работа 3

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1 Цель работы

Приобретение практических навыков по установке и конфигурированию DHCP-сервера.

2 Выполнение лабораторной работы

Зайдём под рутом и установим пакет для настройки dhcp - kea (рис. 2.1).

```
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : mariadb-connector-c-config-3.4.4-2.el10_2.noarch 1/6
  Installing     : mariadb-connector-c-3.4.4-2.el10_2.x86_64      2/6
  Installing     : log4cplus-2.1.1-10.el10.x86_64                3/6
  Installing     : libpq-16.14-1.el10_2.x86_64                   4/6
  Installing     : kea-libs-3.0.3-1.el10_2.x86_64                 5/6
  Running scriptlet: kea-3.0.3-1.el10_2.x86_64                    6/6
  Installing     : kea-3.0.3-1.el10_2.x86_64                     6/6
  Running scriptlet: kea-3.0.3-1.el10_2.x86_64                    6/6

Installed:
  kea-3.0.3-1.el10_2.x86_64
  kea-libs-3.0.3-1.el10_2.x86_64
  libpq-16.14-1.el10_2.x86_64
  log4cplus-2.1.1-10.el10.x86_64
  mariadb-connector-c-3.4.4-2.el10_2.x86_64
  mariadb-connector-c-config-3.4.4-2.el10_2.noarch

Complete!
[root@server.dvkalashnikova.net ~]#
```

Рис. 2.1: Установка kea

Перед изменением конфигурационного файла, на всякий случай сделаем бекап и отредактируем его (рис. 2.2).

```
[root@server.dvkalashnikova.net ~]# cp /etc/kea/kea-dhcp4.conf /etc/kea/kea-dhcp4.conf_$(date +%Y%m%d_%H%M%S)
[root@server.dvkalashnikova.net ~]# nano /etc/kea/kea-dhcp4.conf
```

Рис. 2.2: Бекап конфига

Мы поменяем изначальные данные на свои - изменим доменное имя на собственное, а также поставим ip на ip нашей машины - 192.168.1.1 (рис. 2.3).

```
GNU nano 8.1 /etc/kea/kea-dhcp4.conf
option-data: [
    // When specifying options, you typically need to specify
    // one of (name or code) and data. The full option specification
    // covers name, code, space, csv-format and data.
    // space defaults to "dhcp4" which is usually correct, unless you
    // use encapsulate options. csv-format defaults to "true", so
    // this is also correct, unless you want to specify the whole
    // option value as long hex string. For example, to specify
    // domain-name-servers you could do this:
    // {
    //     "name": "domain-name-servers",
    //     "code": 6,
    //     "csv-format": "true",
    //     "space": "dhcp4",
    //     "data": "192.0.2.1, 192.0.2.2"
    // }
    // but it's a lot of writing, so it's easier to do this instead:
    {
        "name": "domain-name-servers",
        "data": "192.0.2.1, 192.0.2.2"
    },

    // Typically people prefer to refer to options by their names, so they
    // don't need to remember the code names. However, some people like
    // to use numerical values. For example, option "domain-name" uses
    // option code 15, so you can reference to it either by
    // "name": "domain-name" or "code": 15.
    {
        "code": 15,
        "data": "example.org"
    },

    // Domain search is also a popular option. It tells the client to
```

Рис. 2.3: Редактирование конфига

Спустимся ниже и настроим свою подсеть следующим образом (рис. 2.4).

```
// PLANNED SUBNET.
"subnet4": [
    {
        // This defines the whole subnet. Kea will use this information to
        // determine where the clients are connected. This is the whole
        // subnet in your network.

        // Subnet identifier should be unique for each subnet.
        "id": 1,

        // This is mandatory parameter for each subnet.
        "subnet": "192.168.1.0/24",

        // Pools define the actual part of your subnet that is governed
        // by Kea. Technically this is optional parameter, but it's
        // almost always needed for DHCP to do its job. If you omit it,
        // clients won't be able to get addresses, unless there are
        // host reservations defined for them.
        "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],

        // This is one of the subnet selectors. Uncomment the "interface"
        // parameter and specify the appropriate interface name if the DHCPv4
        // server will receive requests from local clients (connected to the
        // same subnet as the server). This subnet will be selected for the
        // requests received by the server over the specified interface.
        // This rule applies to the DORA exchanges and rebinding clients.
        // Renewing clients unicast their messages, and the renewed addresses
        // are used by the server to determine the subnet they belong to.
        // When this parameter is used, the "relay" parameter is typically
        // unused.
        // "interface": "eth0".
```

Рис. 2.4: Настройка подсети

И установим интерфейс для dhcp как eth1 (рис. 2.5).

```
// This file is licensed under the terms of the GNU GPL v2.
"Dhcp4": {
    // Add names of your network interfaces to listen on.
    "interfaces-config": {
        // See section 8.2.4 for more details. You probably want to add just
        // interface name (e.g. "eth0" or specific IPv4 address on that
        // interface name (e.g. "eth0/192.0.2.1").
        "interfaces": ["eth1"]

        // Kea DHCPv4 server by default listens using raw sockets. This ensures
        // all packets, including those sent by directly connected clients
        // that don't have IPv4 address yet, are received. However, if your
        // traffic is always relayed, it is often better to use regular
        // UDP sockets. If you want to do that, uncomment this line:
        // "dhcp-socket-type": "udp"
    },

    // Kea supports control channel, which is a way to receive management
    // commands while the server is running. This is a Unix domain socket that
    // receives commands formatted in JSON, e.g. config-set (which sets new
    // configuration), config-reload (which tells Kea to reload its
    // configuration from file), statistic-get (to retrieve statistics) and many
    // more. For detailed description, see Sections 8.8, 16 and 15.
    "control-socket": {
        "socket-type": "unix",
        "socket-name": "kea4-ctrl-socket"
    }
},
}
```

Рис. 2.5: Установка интерфейса

Загрузим конфиг и убедимся, что нигде нет критических ошибок (рис. 2.6).

```
[root@server.dvkalashnikova.net ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-09-15 15:22:49.752 WARN [kea-dhcp4.dhcp4/14875.140118497761408] DHCP4_SRV_MY_DISABLED_QUEUE_CONTROL disabling dhcp q
ueue control when multi-threading is enabled.
2026-09-15 15:22:49.754 WARN [kea-dhcp4.dhcp4/14875.140118497761408] DHCP4_RESERVATIONS_LOOKUP_FIRST_ENABLED Multi-thread
ing is enabled and host reservations lookup is always performed first.
2026-09-15 15:22:49.755 INFO [kea-dhcp4.dhcp4/14875.140118497761408] DHCP4_SRV_CFGMGR_NEW_SUBNET4 a new subnet has been a
dded to configuration: 192.168.1.0/24 with params: tl=900, t2=1800, valid-lifetime=3600
2026-09-15 15:22:49.759 ERROR [kea-dhcp4.dhcp4/14875.140118497761408] DHCP4_PARSER_FAIL failed to create or run parser for
configuration element subnet4: specified reservation '192.0.2.201' is not within the IPv4 subnet '192.168.1.0/24'
2026-09-15 15:22:49.760 INFO [kea-dhcp4.dhcp4/14875.140118497761408] DHCP4_LEASE_MGR_BACKENDS_REGISTERED the followin
g lease backend types are available: memfile
2026-09-15 15:22:49.760 INFO [kea-dhcp4.dhcp4/14875.140118497761408] DHCP4_HOSTS_BACKENDS_REGISTERED the following host backen
d types are available:
```

Рис. 2.6: Загрузка конфига

Перезагрузим системные демоны (рис. 2.7).

```
[root@server.dvkalashnikova.net ~]# systemctl --system daemon-reload
[root@server.dvkalashnikova.net ~]# systemctl enable kea-dhcp4.service
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp4.service' → '/usr/lib/systemd/system/kea-dhcp4.servi
ce'.
```

Рис. 2.7: Перезагрузка демонов

И слегка отредактируем наш файл с прошлой лабораторной работы в папке fz, добавив запись о dhcp (рис. 2.8).

```
GNU nano 8.1 /var/named/master/fz/dvkalashnikova.net
$TTL 1D
@ IN SOA @ server.dvkalashnikova.net. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

    NS      @
    A       192.168.1.1
    AAAA    ::1
$ORIGIN dvkalashnikova.net.
server A 192.168.1.1
ns A 192.168.1.1
dhcp A 192.168.1.1
```

Рис. 2.8: Редактирование fz

То же самое сделаем с rz (рис. 2.9).

```
GNU nano 8.1 /var/named/master/rz/192.168.1
$TTL 1D
@ IN SOA @ server.dvkalashnikova.net. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

    NS      @
    A       192.168.1.1
    AAAA    ::1
    PTR     server.dvkalashnikova.net
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.dvkalashnikova.net.
1 PTR ns.dvkalashnikova.net.
1 PTR dhcp.dvkalashnikova.net.
```

Рис. 2.9: rz

Перезагрузим сервер ДНС и убедимся, что мы можем пингануть dhcp сервер (рис. 2.10).

```
[root@server.dvkalashnikova.net ~]# systemctl restart named
[root@server.dvkalashnikova.net ~]# ping dhcp.dvkalashnikova.net
PING dhcp.dvkalashnikova.net (192.168.1.1) 56(84) bytes of data:
64 bytes from ns.dvkalashnikova.net (192.168.1.1): icmp_seq=1 ttl=64 time=14.7 ms
64 bytes from ns.dvkalashnikova.net (192.168.1.1): icmp_seq=2 ttl=64 time=3.70 ms
64 bytes from ns.dvkalashnikova.net (192.168.1.1): icmp_seq=3 ttl=64 time=4.14 ms
64 bytes from ns.dvkalashnikova.net (192.168.1.1): icmp_seq=4 ttl=64 time=4.76 ms
64 bytes from ns.dvkalashnikova.net (192.168.1.1): icmp_seq=5 ttl=64 time=7.36 ms
64 bytes from ns.dvkalashnikova.net (192.168.1.1): icmp_seq=6 ttl=64 time=10.9 ms
64 bytes from ns.dvkalashnikova.net (192.168.1.1): icmp_seq=7 ttl=64 time=8.43 ms
```

Рис. 2.10: Пинг dhcp

Теперь настроим firewall и обновим метки selinux (рис. 2.11).

```
[root@server.dvkalashnikova.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns ssh
[root@server.dvkalashnikova.net ~]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 anno-1800 apcupsd
aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc
bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-iv civ
lization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc
dns dns-over-quick dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio
finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ltd freeipa-replication freeipa-trust ftp galera ganglia-cl
ient ganglia-master git gitea gssd grafana gre high-availability http http3 https ident imap imaps iperf2 iperf3 ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogon kpasswd kprop kshell kube-api
kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-n
odeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvir
rt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managelieve matrix mdns memcache minecraft minidln
a mndp mongod mosh moudm mqtg mqtg-tls ms-abt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns ne
tdata-dashboard nfs nfs3 nmap nmap-1813 ntp ntpd ntpd opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole p
lex pcmd pmpoxy pmebapi pmebapi pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp proxy-htt
p ps2link ps3netrvr ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh
rsyncd rtsp salt-master samba samba-client samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submis
sion smtps snmp snmpd snmpd snmpd socks spiderOak-lansync spotify-sync squid ssh statrsv steam-lan-transf
er steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui sync
thing-relay synergy syslog syslog-ng syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission
-client turn turns upnp-client vdsms vnc-server vrrp warpinator wbm-http wbm-https wireguard ws-discovery ws-discovery-cl
ient ws-discovery-host ws-discovery-tcp ws-discovery-udp wssd wssd-http wsmann wsmans xdmcp xmpp-bosh xmpp-client xmpp-loc
al xmpp-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=dhcp
success
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=dhcp --permanent
success
[root@server.dvkalashnikova.net ~]# restorecon -vR /etc
[root@server.dvkalashnikova.net ~]# restorecon -vR /var/named
[root@server.dvkalashnikova.net ~]# restorecon -vR /var/lib/kea/
```

Рис. 2.11: firewall и selinux

Убедимся по логам, что сервер ДНС работает и не выдаёт ошибок (рис. 2.12).

```
[dvkalashnikova@server.dvkalashnikova.net ~]$ sudo tail -f /var/log/messages
[sudo] password for dvkalashnikova:
Sep 15 15:36:55 server systemd-coredump[17295]: Process 17290 (VBoxClient) of user 1001 dumped core.#012#012Module /opt/VB
oxGuestAdditions-7.2.16/bin/VBoxClient without build-id.#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64#012Mod
ule libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module
libffi.so.8 from rpm libffi-3.4.4-10.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.24.0-1.el10.x86_64#01
2Stack trace of thread 17293:#012#0 0x00000000041e7eb n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x1e7eb)#012#
1 0x0000000000041e76 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x1e76)#012#2 0x0000000000458b2c n/a (/opt/VB
oxGuestAdditions-7.2.16/bin/VBoxClient + 0x58b2c)#012#3 0x000000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCl
ient + 0x3b900)#012#4 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#5 0x00007fcec496343c __clone3 (libc.so.6
+ 0x10643c)#012#12Stack trace of thread 17290:#012#0 0x00007fcec496125d syscall (libc.so.6 + 0x10425d)#012#1 0x0000000
00043a812 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x3a812)#012#2 0x0000000000458786 n/a (/opt/VBoxGuestAddit
ions-7.2.16/bin/VBoxClient + 0x58786)#012#3 0x00000000004056d8 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x56d
8)#012#4 0x00007fcec488758e __libc_start_call_main (libc.so.6 + 0x2a58e)#012#5 0x00007fcec4887649 __libc_start_main@GLI
BC.2.34 (libc.so.6 + 0x2a649)#012#6 0x00000000004044fa n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x44fa)#012EL
F object binary architecture: AMD x86_64
Sep 15 15:36:55 server systemd[1]: systemd-coredump[446-17294-0.service: Deactivated successfully.
Sep 15 15:37:00 server kernel: traps: VBoxClient[17307] trap int3 ip:41e7eb sp:7fceb6235cd0 error:0 in VBoxClient[1e7eb,40
0000c8000]
Sep 15 15:37:00 server systemd-coredump[17308]: Process 17304 (VBoxClient) of user 1001 terminated abnormally with signal
5/TRAP, processing...
Sep 15 15:37:00 server systemd[1]: Started systemd-coredump[447-17308-0.service - Process Core Dump (PID 17308/UID 0).
Sep 15 15:37:00 server systemd-coredump[17309]: Process 17304 (VBoxClient) of user 1001 dumped core.#012#012Module /opt/VB
oxGuestAdditions-7.2.16/bin/VBoxClient without build-id.#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64#012Mod
ule libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module
libffi.so.8 from rpm libffi-3.4.4-10.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.24.0-1.el10.x86_64#01
2Stack trace of thread 17307:#012#0 0x000000000041e7eb n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x1e7eb)#012#
1 0x0000000000041e76 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x1e76)#012#2 0x0000000000458b2c n/a (/opt/VB
oxGuestAdditions-7.2.16/bin/VBoxClient + 0x58b2c)#012#3 0x000000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCl
ient + 0x3b900)#012#4 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#5 0x00007fcec496343c __clone3 (libc.so.6
+ 0x10643c)#012#12Stack trace of thread 17305:#012#0 0x00007fcec496125d syscall (libc.so.6 + 0x10425d)#012#1 0x0000000
00043af60 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x3af60)#012#2 0x000000000045930b n/a (/opt/VBoxGuestAddit
ions-7.2.16/bin/VBoxClient + 0x5930b)#012#3 0x000000000043b99a n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x3b9
9a)#012#4 0x0000000000458b2c n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x58b2c)#012#5 0x000000000043b900 n/a
(/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x3b900)#012#6 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012
#7 0x00007fcec48f28bf __libc_start_call_main (libc.so.6 + 0x2a58e)#012#12Stack trace of thread 17304:#012#0 0x00007fcec48f28bf start
thread (libc.so.6 + 0x958bf)#012#1 0x0000000000458b2c n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x58b2c)#012#2 0x000000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x3b900)#012#3 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#4 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#5 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#6 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#7 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#8 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#9 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#10 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#11 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#12 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#13 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#14 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#15 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#16 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+ 0x958bf)#012#98 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#99 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#100 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#101 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#102 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#103 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#104 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#105 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#106 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#107 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#108 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#109 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#110 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#111 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#112 0x00007fcec48f28bf start_thread (libc.so.6 + 0x958bf)#012#113 0x00007fcec48f28bf start_thread (libc.so.6 + 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```

```
[root@server.dvkalashnikova.net ~]# systemctl start kea-dhcp4.service
[root@server.dvkalashnikova.net ~]#
```

Рис. 2.13: Запуск dhcp

Далее убедимся в том, что в нашей папке клиента в vagrant представлен скрипт следующего содержания для настройки сети, берущей свой ip по dhcp (рис. 2.14).

```
#!/bin/bash

echo "Provisioning script $0"

nmcli connection modify "System eth1" ipv4.gateway "192.168.1.1"
nmcli connection up "System eth1"

nmcli connection modify eth0 ipv4.never-default true
nmcli connection modify eth0 ipv6.never-default true

nmcli connection down eth0
nmcli connection up eth0

# systemctl restart NetworkManager
```

Рис. 2.14: Скрипт для клиента

В Vagrantfile мы убедимся, что этот скрипт прописан для запуска (рис. 2.15).

```
client.vm.provision "client dummy",
  type: "shell",
  preserve_order: true,
  path: "provision/client/01-dummy.sh"

client.vm.provision "client routing",
  type: "shell",
  preserve_order: true,
  run: "always",
  path: "provision/client/01-routing.sh"
```

Рис. 2.15: Vagrantfile

Когда приготовления завершены, мы можем запустить клиент (рис. 2.16).

```
C:\work\dvkalashnikova\code\vagrant> vagrant up client --provision
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Clearing any previously set forwarded ports...
```

Рис. 2.16: Запуск клиента

Зайдя в клиент, через ifconfig убедимся, что айпи был получен с сервера. Это так, айпи назначился как 192.168.1.30 (рис. 2.17).

```
[dvkalashnikova@client ~]$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:a00:27ff:fe64:5285 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:64:52:85 txqueuelen 1000 (Ethernet)
    RX packets 977 bytes 146887 (143.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 886 bytes 135693 (132.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
```

Рис. 2.17: ifconfig

Информация о назначении айпи также хранится в файле /var/lib/kea/kea-leases4.csv на сервере (рис. 2.18).

GNU nano 8.1	/var/lib/kea/kea-leases4.csv	Modified
address, hwaddr, client_id, valid_lifetime, expire, subnet_id, fqdn_fwd, fqdn_rev, hostname, state, user_c		
192.168.56.30, 08:00:27:8e:21:48, 01:08:00:27:8e:21:48, 3600, 1758375351, 1, 0, 0, client, 0, 0		

Рис. 2.18: Таблица с назначениями

Теперь создадим ключ sha512 и убедимся в том, что он создан (рис. 2.19).

```
[root@server.dvkalashnikova.net ~]# mkdir -p /etc/named/keys
[root@server.dvkalashnikova.net ~]# tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key
[root@server.dvkalashnikova.net ~]# cat /etc/named/keys/dhcp_updater.key
key "DHCP_UPDATER" {
    algorithm hmac-sha512;
    secret "ApIQM0/0eMfuPLLVKBhuHzcE0JFVBpHB4GwdarD8XIhYN7rj1jcJIqqb7LYOpCXXd0v2qxMMqtHjEWLyPbo7jQ==";
};
```

Рис. 2.19: Ключ sha512

Этот ключ добавим в /etc/named.conf (рис. 2.20).

```

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
include "/etc/named/dvkalashnikova.net";
include "/etc/named/keys/dhcp_updater.key";

```

Рис. 2.20: Добавление ключа

Обновим файл `/etc/named/dvkalashnikova.net`, добавив туда `dhcp` (рис. 2.21).

```

GNU nano 8.1 /etc/named/dvkalashnikova.net Modif
// named.rfc1912.zones:
//
// Provided by Red Hat caching-nameserver package
//
// ISC BIND named zone configuration for zones recommended by
// RFC 1912 section 4.1 : localhost TLDs and address zones
// and https://tools.ietf.org/html/rfc6303
// (c)2007 R W Franks
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
// Note: empty-zones-enable yes; option is default.
// If private ranges should be forwarded, add
// disable-empty-zone "."; into options
//
zone "dvkalashnikova.net" IN {
    type master;
    file "master/fz/dvkalashnikova.net";
    update-policy {
        grant DHCP_UPDATER wildcard *.user.net A DHCID;
    };
};

zone "1.168.192.in-addr.arpa" IN {
    type master;
    file "master/rz/192.168.1";
    update-policy {
        grant DHCP_UPDATER wildcard *.1.168.192.in-addr.arpa PTR DHCID;
    };
};

```

Рис. 2.21: Обновление файла

Проверим корректность конфига на синтаксис и перезапустим DNS службу, а также создадим файл с ключом (рис. 2.22).


```

GNU nano 8.1 /etc/kea/kea-dhcp-ddns.conf Modified
{
  "DhcpDdns": {
    {
      "ip-address": "127.0.0.1",
      "port": 53001,
      "control-socket": {
        "socket-type": "unix",
        "socket-name": "/run/kea/kea-ddns-ctrl-socket"
      },
      <?include "/etc/kea/tsig-keys.json"?>
      "forward-ddns": {
        "ddns-domains": [
          {
            "name": "dvkalashnikova.net.",
            "key-name": "DHCP_UPDATER",
            "dns-servers": [
              { "ip-address": "192.168.1.1" }
            ]
          }
        ],
        "reverse-ddns": {
          "ddns-domains": [
            {
              "name": "1.168.192.in-addr.arpa.",
              "key-name": "DHCP_UPDATER",
              "dns-servers": [
                { "ip-address": "192.168.1.1" }
              ]
            }
          ]
        },
        "loggers": [
          {
            "name": "kea-dhcp-ddns",
            "output_options": [

```

Рис. 2.25: Изменение конфигурации

Предоставим этот файл во владение системному пользователю, а также загрузим эту конфигурацию, и убедимся, что она загружена успешно. После этого перезапустим ddns службу (рис. 2.26).

```

[root@server.dvkalashnikova.net ~]# systemctl enable --now kea-dhcp-ddns.service
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service' → '/usr/lib/systemd/system/kea-dhcp-ddns.service'.
[root@server.dvkalashnikova.net ~]# systemctl status kea-dhcp-ddns.service
● kea-dhcp-ddns.service - Kea DHCP-DDNS Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 17:53:42 UTC; 57s ago
   Invocation: e889dec48149f6b6eb89b5a3a68b3e
     Docs: man:kea-dhcp-ddns(8)
    Main PID: 38987 (kea-dhcp-ddns)
      Status: "Dispatching packets..."
        Tasks: 5 (limit: 35588)
      Memory: 2.3M (peak: 6.9M)
         CPU: 158ms
    CGroup: /system.slice/kea-dhcp-ddns.service
            └─38987 /usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf

Sep 15 17:53:42 server.dvkalashnikova.net systemd[1]: Starting kea-dhcp-ddns.service - Kea DHCP-DDNS Server...
Sep 15 17:53:42 server.dvkalashnikova.net kea-dhcp-ddns[38987]: 2026-09-15 17:53:42.946 INFO [kea-dhcp-ddns.dctl]
Sep 15 17:53:42 server.dvkalashnikova.net kea-dhcp-ddns[38987]: INFO COMMAND_ACCEPTOR_START Starting to accept
Sep 15 17:53:42 server.dvkalashnikova.net kea-dhcp-ddns[38987]: INFO DCTL_CONFIG_COMPLETE server has completed
Sep 15 17:53:42 server.dvkalashnikova.net kea-dhcp-ddns[38987]: INFO DHCP_DDNS_STARTED Kea DHCP-DDNS server ver

```

Рис. 2.26: Перезапуск ddns

Теперь добавим информацию о ddns в наш файл с конфигурацией dhcp (рис. 2.27).

```

GNU nano 8.1 /etc/kea/kea-dhcp-ddns.conf
{
  "DhcpDdns": {
    {
      "ip-address": "127.0.0.1",
      "port": 53001,
      "control-socket": {
        "socket-type": "unix",
        "socket-name": "/var/run/kea/kea-ddns-ctrl-socket"
      },
      <?include "/etc/kea/tsig-keys.json"?>
      "forward-ddns": {
        "ddns-domains": [
          {
            "name": "dvkalashnikova.net.",
            "key-name": "DHCP_UPDATER",
            "dns-servers": [
              { "ip-address": "192.168.1.1" }
            ]
          }
        ]
      },
      "reverse-ddns": {
        "ddns-domains": [
          {
            "name": "1.168.192.in-addr.arpa.",
            "key-name": "DHCP_UPDATER",
            "dns-servers": [
              { "ip-address": "192.168.1.1" }
            ]
          }
        ]
      },
      "loggers": [
        {
          "name": "kea-dhcp-ddns",
          "output_options": [
            {
              "output": "stdout",
              "pattern": "%-5p %m\n"
            }
          ],
          "severity": "INFO",
          "debuglevel": 0
        }
      ]
    }
  }
}

```

Рис. 2.27: Добавление информации о ddns

Вновь загрузим конфигурацию и перезапустим службу dhcp (рис. 2.28).

```

[root@server.dvkalashnikova.net ~]# systemctl restart kea-dhcp4.service
[root@server.dvkalashnikova.net ~]# systemctl status kea-dhcp4.service
● kea-dhcp4.service - Kea DHCPv4 Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-15 18:15:44 UTC; 17s ago
     Invocation: ee4b4e727b5a40c1b428191240546c56
       Docs: man:kea-dhcp4(8)
    Main PID: 43238 (kea-dhcp4)
      Status: "Dispatching packets..."
        Tasks: 9 (limit: 35588)
      Memory: 2.6M (peak: 7M)
         CPU: 184ms
    CGroup: /system.slice/kea-dhcp4.service
            └─43238 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

Sep 15 18:15:44 server.dvkalashnikova.net systemd[1]: Starting kea-dhcp4.service - Kea DHCPv4 Server...
Sep 15 18:15:44 server.dvkalashnikova.net kea-dhcp4[43238]: 2026-09-15 18:15:44.658 INFO [kea-dhcp4.dhcp4/43238]
Sep 15 18:15:44 server.dvkalashnikova.net kea-dhcp4[43238]: 2026-09-15 18:15:44.678 WARN [kea-dhcp4.dhcp4/43238]
Sep 15 18:15:44 server.dvkalashnikova.net kea-dhcp4[43238]: 2026-09-15 18:15:44.679 INFO [kea-dhcp4.commands/43238]
Sep 15 18:15:44 server.dvkalashnikova.net systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.

```

Рис. 2.28: Перезапуск службы с применением изменений

Теперь на клиенте перезапустим интернет, чтобы обновить данные (рис. 2.29).

```
[dvkalashnikova@client ~]$ nmcli connection down eth1
Connection 'eth1' successfully deactivated (D-Bus active path: /org/freedesktop/
NetworkManager/ActiveConnection/7)
[dvkalashnikova@client ~]$ nmcli connection up eth1
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkMa
nager/ActiveConnection/8)
[dvkalashnikova@client ~]$
```

Рис. 2.29: Обновление данных

Теперь через dig получим информацию о нашем сервере (рис. 2.30).

```
;; no servers could be reached
[dvkalashnikova@client ~]$ dig @192.168.56.1 client.dvkalashnikova.net
;; communications error to 192.168.56.1#53: timed out
;; communications error to 192.168.56.1#53: timed out
;; communications error to 192.168.56.1#53: timed out

; <<>> DiG 9.18.33 <<>> @192.168.56.1 client.dvkalashnikova.net
; (1 server found)
;; global options: +cmd
;; no servers could be reached
[dvkalashnikova@client ~]$ dig @192.168.56.1 client.dvkalashnikova.net
;; communications error to 192.168.56.1#53: timed out
;; communications error to 192.168.56.1#53: timed out
;; communications error to 192.168.56.1#53: timed out

; <<>> DiG 9.18.33 <<>> @192.168.56.1 client.dvkalashnikova.net
; (1 server found)
;; global options: +cmd
;; no servers could be reached
[dvkalashnikova@client ~]$
```

Рис. 2.30: dig

Теперь переместим данные созданных ранее конфигураций в вагрант, после чего создадим скрипт (рис. 2.31).

```
[root@server.dvkalashnikova.net dns]# cd /vagrant/provision/server
[root@server.dvkalashnikova.net server]# ouch dhcp.sh
ouch: ouch: command not found...
[root@server.dvkalashnikova.net server]# touch dhcp.sh
[root@server.dvkalashnikova.net server]# chmod +x dhcp.sh
```

Рис. 2.31: Перенос конфигурации

В скрипте напишем алгоритм настройки dhcp (рис. 2.32).

```
GNU nano 8.1                                dhcp.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install kea
echo "Copy configuration files"
cp -R /vagrant/provision/server/dhcp/etc/kea/* /etc/kea/
echo "Fix permissions"
chown -R kea:kea /etc/kea
chmod 640 /etc/kea/tsig-keys.json
restorecon -vR /etc
restorecon -vR /var/lib/kea
echo "Configure firewall"
firewall-cmd --add-service dhcp
firewall-cmd --add-service dhcp --permanent
echo "Start dhcpd service"
systemctl --system daemon-reload
```

Рис. 2.32: Скрипт vagrant

3 Выводы

В результате выполнения работы были получены навыки настройки dhcp.